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-: Assignment no. 5: -

Write cursor code using control structures and exception handling.

1. Cursor (row-by-row)

2. Set-based (without cursor)

Problem Statement

In an Employee Appraisal System:

· Increase salary based on rating

· Also assign bonus category:

o Rating 5 → "Excellent"

o Rating 4 → "Good"

o Rating 3 → "Average"

o Else → "Needs Improvement"

```
mysql> USE Employeedb;
Database changed
mysql> CREATE TABLE employees (
    ->     emp_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     salary INT,
    ->     rating INT,
    ->     bonus_category VARCHAR(30)
    -> );
ERROR 1050 (42S01): Table 'employees' already exists
```

```
mysql> INSERT INTO employees VALUES
    -> (101, 'Amit', 50000, 5, NULL),
    -> (102, 'Neha', 40000, 4, NULL),
    -> (103, 'Raj', 30000, 3, NULL),
    -> (104, 'Simran', 35000, 2, NULL);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE appraisal_cursor()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE v_id INT;
    ->     DECLARE v_salary INT;
    ->     DECLARE v_rating INT;
    ->     DECLARE v_category VARCHAR(30);
    ->
    ->     DECLARE emp_cursor CURSOR FOR
    ->         SELECT emp_id, salary, rating FROM employees;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN emp_cursor;
    ->
    ->     read_loop: LOOP
    ->         FETCH emp_cursor INTO v_id, v_salary, v_rating;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         -- Logic
    ->         IF v_rating = 5 THEN
    ->             SET v_salary = v_salary + 10000;
    ->             SET v_category = 'Excellent';
    ->
    ->         ELSEIF v_rating = 4 THEN
    ->             SET v_salary = v_salary + 5000;
    ->             SET v_category = 'Good';
    ->
    ->         ELSEIF v_rating = 3 THEN
    ->             SET v_salary = v_salary + 2000;
    ->             SET v_category = 'Average';
    ->
    ->         ELSE
    ->             SET v_category = 'Needs Improvement';
    ->         END IF;
    ->
    ->         UPDATE employees
    ->         SET salary = v_salary,
```

```
    ->
    ->         UPDATE employees
    ->         SET salary = v_salary,
    ->             bonus_category = v_category
    ->         WHERE emp_id = v_id;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE emp_cursor;
    ->
    -> END$$
ERROR 1304 (42000): PROCEDURE appraisal_cursor already exists
mysql>
mysql> DELIMITER ;
mysql> CALL appraisal_cursor();
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM employees;
+--------+--------+--------+--------+------------------+
| emp_id | name   | salary | rating | bonus_category   |
+--------+--------+--------+--------+------------------+
|    101 | Amit   |  60000 |      5 | Excellent        |
|    102 | Neha   |  45000 |      4 | Good             |
|    103 | Raj    |  32000 |      3 | Average          |
|    104 | Simran |  35000 |      2 | Needs Improvement |
+--------+--------+--------+--------+------------------+
4 rows in set (0.00 sec)
```

Problem Statement

For each employee:

· If salary > 50000 → deduct 10% tax

· Insert tax record into another table

· Maintain running total of tax collected Here:

· Each row updates another table

· Also depends on previous total (running sum) This becomes difficult (or messy) in pure set-based SQL.

```
mysql> CREATE TABLE employees (
    ->     emp_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     salary INT
    -> );
ERROR 1050 (42S01): Table 'employees' already exists
mysql> CREATE TABLE tax_records (
    ->     emp_id INT,
    ->     tax_amount DECIMAL(10,2),
    ->     total_tax_collected DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO employees (emp_id, name, salary) VALUES
    -> (101, 'Amit', 60000),
    -> (102, 'Neha', 45000),
    -> (103, 'Raj', 70000),
    -> (104, 'Simran', 30000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE calculate_tax()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE v_id INT;
    ->     DECLARE v_salary INT;
    ->     DECLARE v_tax DECIMAL(10,2);
    ->     DECLARE total_tax DECIMAL(10,2) DEFAULT 0;
    ->
    ->     DECLARE emp_cursor CURSOR FOR
    ->         SELECT emp_id, salary FROM employees;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN emp_cursor;
    ->
    ->     read_loop: LOOP
    ->         FETCH emp_cursor INTO v_id, v_salary;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         -- Tax logic
    ->         IF v_salary > 50000 THEN
    ->             SET v_tax = v_salary * 0.10;
    ->         ELSE
    ->             SET v_tax = 0;
    ->         END IF;
    ->
    ->         -- Running total
    ->         SET total_tax = total_tax + v_tax;
    ->
    ->         -- Insert into tax table
    ->         INSERT INTO tax_records(emp_id, tax_amount, total_tax_collected)
    ->         VALUES (v_id, v_tax, total_tax);
    ->
    ->     END LOOP;
    ->
    ->     CLOSE emp_cursor;
    ->
    -> END$$
```

```
ERROR 1304 (42000): PROCEDURE calculate_tax already exists
mysql>
mysql> DELIMITER ;
mysql> CALL calculate_tax();
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM tax_records;
+--------+------------+---------------------+
| emp_id | tax_amount | total_tax_collected |
+--------+------------+---------------------+
|    101 |    6000.00 |             6000.00 |
|    102 |       0.00 |             6000.00 |
|    103 |    7000.00 |            13000.00 |
|    104 |       0.00 |            13000.00 |
+--------+------------+---------------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT MAX(total_tax_collected) AS total_tax
    -> FROM tax_records;
+-----------+
| total_tax |
+-----------+
|  13000.00 |
+-----------+
1 row in set (0.01 sec)
```